

MA 561: Fluidics and Microfluidics

Course Code	MA 561
Course Name	Fluidics and Microfluidics
L-T-P-C	3-0-2-4
Prerequisites	MA 220 (PDE), MA 221 (Numerical Analysis) or equivalent
Intended for	BTech Maths and Computing 3rd year, 4th year, MSc, PhD, Elective for other disciplines.
Distribution	Elective
Mutual Exclusion	ME 507
Approval	61st BoA (Jan 2026)

Preamble

The objective of this course is to provide students with a thorough understanding of fluid dynamics from fundamentals to microscale applications. Emphasis is placed on developing computational proficiency using COMSOL Multiphysics, a modern simulation platform widely used in research and industry, to model and analyze microfluidic, interfacial, and electrohydrodynamic flow phenomena.

Course Modules

- Fundamentals of Fluid Dynamics:** Review of fluid properties: viscosity, density, surface tension, compressibility; Control volume and differential formulations; conservation of mass and momentum; Reynolds transport theorem; Navier–Stokes and Energy equations (steady and unsteady forms); Pressure-driven flows: exact solutions for Couette and Poiseuille flows; Stokes drag on a sphere, creeping flow, laminar flow approximations. (12 Hours)
 Lab: Basics of Finite Difference; Numerical modelling of 1D pressure driven flows such as Couette flow, Poiseuille flow, Hagen-Poiseuille flow.
- Introduction to Microfluidics and its applications:** Origin, definition, motivation, benefits, challenges, and applications; Physics of miniaturization, dimensional analysis, and scaling laws; Continuum hypothesis; Boundary conditions: slip theory, entrance effects, low Re regimes; Applications: Drug delivery and diagnostics; Biosensing and lab-on-a-chip systems. (10 Hours)
 Lab: An overview of solving Navier-Stokes via any discrete numerical methods (understanding the sensitivity of pressure); Basics of COMSOL Multiphysics: model building, geometry, physics, mesh, study, post processing (result plotting 1D, 2D, and 3D). Fluid flow module- Flow past a sphere.
- Capillary Flows:** Surface tension and interfacial energy; Young–Laplace equation, contact angle, capillary rise and capillary length; Interfacial boundary conditions; Marangoni effect; capillary flows. (10 Hours)
 Lab: COMSOL: Laminar two-phase flow: (level set, phase field) Young–Laplace droplet on a flat substrate (2D axisymmetric); Capillary rise in a tube (2D axisymmetric); Marangoni-driven flow along an interface (thermocapillary and solutal marangoni).
- Electrohydrodynamics & Microchannel Flows:** Fundamentals of electrohydrodynamics and species transport; Poisson–Boltzmann equation, Debye layer, thin EDL approximation; Electroosmosis: ideal Electroosmotic Flow (EOF), EOF with back pressure; EO devices- micropumps, micromixers, microvalve; EOF of power-law fluids. (10 Hours)
 Lab: COMSOL: Lid driven cavity combined with mass transport and heat transport, Electric field in a dielectric or leaky dielectric fluid, Electrostatic deformation of a droplet (leaky dielectric model), Electroosmotic flow in a microchannel, Electroosmotic flow in a microchannel connected reservoir, EOF of power-law fluids.

Laboratory Modules

Mentioned with the corresponding theory units.

Text Books

- Bruus, Henrik. *Theoretical microfluidics*. Vol. 18. Oxford university press, 2007.
- White, Frank M. *Fluid mechanics*. International Edition, McGraw-Hill (2015).

References

- Kirby, B. J. – *Micro- and Nanoscale Fluid Mechanics: Transport in Microfluidic Devices*, Cambridge University Press.
- Tabeling, P. – *Introduction to Microfluidics*, Oxford University Press.
- Nguyen, N.-T. & Wereley, S. T. – *Fundamentals and Applications of Microfluidics*, Artech House.
- Batchelor, George Keith. *An introduction to fluid dynamics*. Cambridge university press, 2000.
- Chaurasia, Ashish S. *Computational fluid dynamics and comsol Multiphysics: A step-by-step approach for chemical engineers*. Apple Academic Press, 2021.
- Multiphysics, COMSOL. “Introduction to comsol multiphysics®.” *COMSOL Multiphysics, Burlington, MA, accessed Feb 9, 2018* (1998): 32.